

THERMAL METRIC ⁽¹⁾		TPS63900	UNIT
		DSK Package (WSON)	
		10 PINS	
Ψ_{JB}	Junction-to-board characterization parameter	31.0	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	10.0	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

6.5 Electrical Characteristics

Over operating junction temperature range and recommended supply voltage range (unless otherwise noted). Typical values are at $V_I = 3.0\text{ V}$, $V_O = 2.5\text{ V}$ and $T_J = 25^\circ\text{C}$ (unless otherwise noted).

PARAMETER			TEST CONDITIONS	MIN	TYP	MAX	UNIT
SUPPLY							
	Quiescent current into VIN		V _(EN) = 3 V, no load, not switching, "unlimited" current setting		0.075	1	μA
	Shutdown current into VIN		V _(EN) = 0 V		60		nA
V _{IT+(UVLO)}	Positive-going UVLO threshold voltage			1.73	1.75	1.77	V
V _{hys(UVLO)}	UVLO threshold voltage hysteresis			90	100	110	mV
V _{IT+(POR)}	Positive-going POR threshold voltage			1.37		1.74	V
I/O SIGNALS							
V _{IH}	High-level input voltage (EN, SEL)					1.2	V
V _{IL}	Low-level input voltage (EN, SEL)			0.4			V
	Input current (EN, SEL)		V _(EN) , V _(SEL) = 1.8 V or 0 V. T _J = 25°C		±1	±10	nA
POWER SWITCH							
r _{DS(on)}	On-state resistance	Q1	V _I = 3 V, V _O = 5 V, test current = 1 A		155		mΩ
		Q2	V _I = 3 V, V _O = 3 V, test current = 1 A		110		
		Q3	V _I = 3 V, V _O = 3 V, test current = 1 A		110		
		Q4	V _I = 5 V, V _O = 3 V, test current = 1 A		155		
CURRENT LIMIT							
	Peak current limit during Startup (Q1)		V _I = 3.6 V, unlimited current limit setting	0.35		0.83	A
	Peak current limit (Q1)		V _I = 1.8 V, V _O = 3.6 V, unlimited current limit setting	1.33	1.45	1.6	A
V _I = 3.6 V, V _O = 3.3 V, 100-mA current limit setting			0.15	0.29	0.51		
	Average input current limit		T _J = −40°C to 85°C	1-mA setting	1		mA
2.5-mA setting				2.5			
5-mA setting				5			
10-mA settting				10			
25-mA setting				25			
50-mA setting				50			
100-mA setting				100			
OUTPUT							
	Output voltage DC accuracy		I _O = 1 mA, C _{O(eff)} = 10 μF, L _(eff) = 2.2 μH			±1.5	%

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PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
CONTROL						
	Internal reference resistor		33			kΩ
R_CFG	R2D setting #0		0	0.1		kΩ
	R2D setting #1		−3%	0.511	+3%	
	R2D setting #2		−3%	1.15	+3%	
	R2D setting #3		−3%	1.87	+3%	
	R2D setting #4		−3%	2.74	+3%	
	R2D setting #5		−3%	3.83	+3%	
	R2D setting #6		−3%	5.11	+3%	
	R2D setting #7		−3%	6.49	+3%	
	R2D setting #8		−3%	8.25	+3%	
	R2D setting #9		−3%	10.5	+3%	
	R2D setting #10		−3%	13.3	+3%	
	R2D setting #11		−3%	16.2	+3%	
	R2D setting #12		−3%	20.5	+3%	
	R2D setting #13		−3%	24.9	+3%	
	R2D setting #14		−3%	30.1	+3%	
	R2D setting #15		−3%	36.5	+3%	
PROTECTION FEATURES						
	Thermal shutdown threshold temperature		140	150	160	°C
	Thermal shutdown hysteresis		15	20	25	°C
TIMING PARAMETERS						
t_d(POR)	POR signal delay after reaching POR threshold		3.8			ms
t_d(EN)	Delay between a rising edge on the EN pin and the start of the output voltage ramp	Supply voltage stable before EN pin goes high	1.5			ms
t_w(SS)	Soft-start step duration	V_O > 1.8 V	100	125	150	μs
t_d(SEL)	Delay between a change in the state of the SEL pin and the first step change in the output voltage		30 40			μs
t_w(DVS)	Dynamic voltage scaling step duration		100	125	150	μs
t_d(RESTART)	Restart delay after protection		10 11			ms